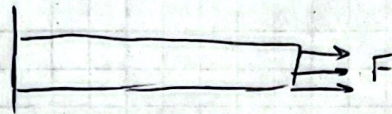


Values Chosen:

Since Cross-Sectional Area is Circular:

- Diameter (d) = 0.1382 m
- Force (F) = 300 lb_f
- Young's Modulus (E) = $8.5 \cdot 10^6$ psi
- Deflection (δ) = 0.009 m

FBD:



Values Given:

- $\Delta L_{\max}$ (Deflection) = 0.009 m
- σ_{yield} of Aluminum = 40 ksi

Elongation Equation: $\Delta L = \frac{PL}{AE}$ (eq 1) $(P=F)$

Solving For Length (L): $L = \frac{\Delta L(AE)}{P}$

$$L = \frac{(0.009 \text{ m}) \left(\frac{\pi}{4} (0.1382 \text{ m})^2 \right) (8.5 \cdot 10^6 \text{ psi})}{300 \text{ lb}_f}$$

$$\boxed{L = 3.825 \text{ m}}$$

Cross-Sectional Area (A) = $\frac{\pi}{4} (d)^2$

$$A = \frac{\pi}{4} (0.1382 \text{ m})^2 = \boxed{0.015 \text{ m}^2}$$

Stress Values:

$$\sigma_{allowable\ calculated} = \frac{F}{A} = \frac{(300\text{ lbf})}{0.015\text{ in}^2} = \frac{20000\text{ psi}}{(= 20\text{ ksi})}$$

$$FOS_{calculated} = \frac{\sigma_{yield}}{\sigma_{allow\ calculated}} = \frac{(40\text{ ksi})}{(20\text{ ksi})} = \boxed{2}$$

Deflection Map:

$$\Delta L_{calculated} = 0.009\text{ in}$$

$$\Delta L_{CAD} = 8.771 \cdot 10^{-3}\text{ in} = 0.008771\text{ in}$$

$$\begin{aligned} \text{Percent error: } & \left| \frac{\text{Calculated} - \text{Actual}}{\text{Actual}} \right| \cdot 100 \\ &= \left| \frac{(0.008771\text{ in}) - (0.009\text{ in})}{(0.009\text{ in})} \right| \cdot 100 \\ &= \boxed{2.54\%} \end{aligned}$$

Von Mises Map:

$$\sigma_{allowable\ calculated} = 20\text{ ksi}$$

$$\sigma_{allowable\ CAD} = 20.42\text{ ksi}$$

$$\begin{aligned} \text{Percent error: } & \left| \frac{\text{Calculated} - \text{Actual}}{\text{Actual}} \right| \cdot 100 \\ &= \left| \frac{(20\text{ ksi}) - (20.42\text{ ksi})}{(20.42\text{ ksi})} \right| \cdot 100 \\ &= \boxed{2.06\%} \end{aligned}$$

Stress Values (cont.)

Factor of Safety (FOS) Max:

$$FOS_{\text{calculated}} = \frac{\sigma_{\text{yield}}}{\sigma_{\text{allowable}}} = \frac{40 \text{ ksi}}{20 \text{ ksi}} = \boxed{2}$$

$$\text{Min } FOS_{\text{CAD}} = \boxed{2}$$

Since $\sigma_{\text{yield}} > \sigma_{\text{allowable}}$:

Bar will not fail under load force (F)

Pin Hole:

$$\Delta L_{\text{calculated}} = \frac{F(L)}{AE} = \frac{(300 \text{ lbf})(3.825 \text{ in})}{(0.015 \text{ in}^2)(9.5 \times 10^6 \text{ psi})}$$

$$\Delta L_{\text{calculated}} = \Delta L_{\text{given}} = 0.0007 \text{ in}$$

ALL Percent errors show that the values are essentially the same

Stress Concentration Factor (K_T) For Hole in Flat bar in Tension:

"Substantial" pin hole $\rightarrow d_{\text{pin}} = 2d = 2(0.1382)$

$$\frac{d}{d_{\text{pin}}} = \frac{d}{W} = \frac{0.1382 \text{ in}}{0.2764 \text{ in}} = \boxed{0.5} \quad \boxed{W = d_{\text{pin}} = 0.2764 \text{ in}}$$

$$\sigma_{\text{allowable CAD}} (K_T) = \sigma_{\text{peak}}$$

$$\frac{d}{W} = 0.5 \rightarrow K_T \approx 2.16 \rightarrow (20.42 \text{ ksi})(2.16) = \boxed{44.11 \text{ ksi} = \sigma_{\text{peak}}}$$

$$FOS_{\text{pin}} = \frac{\sigma_{\text{yield}}}{\sigma_{\text{peak}}} = \frac{40 \text{ ksi}}{44.11 \text{ ksi}} = \boxed{0.91}$$

Since $FOS_{\text{pin}} < 1 \rightarrow$ Bar will fail with pin hole